



Neurological Emergency

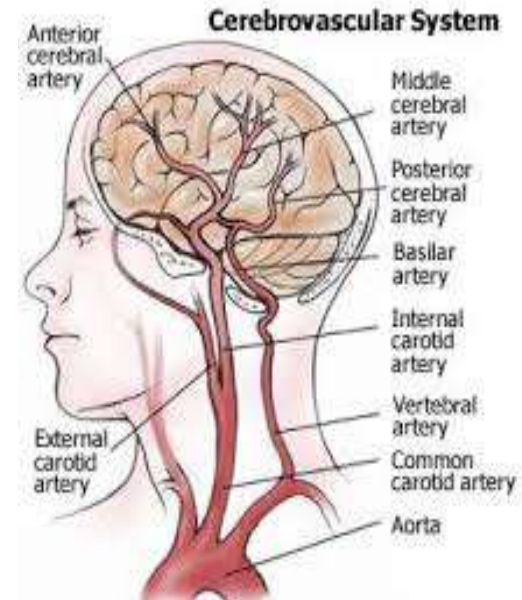
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Learning Outcomes

- **By the end of this presentation, students should be able to:**
 - **Define Cerebrovascular Stroke.**
 - **Define epilepsy.**
 - **Discuss classification and management of epilepsy .**
 - **Manage status epilepticus.**

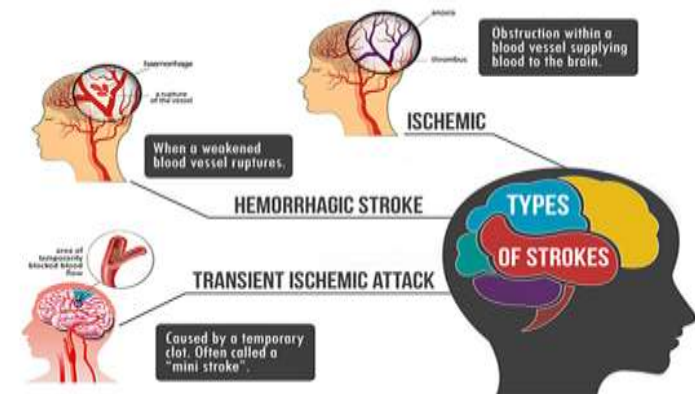
Cerebrovascular stroke

- Blood enters the brain through:
 - Two carotid and two vertebral arteries (supply specific area of the brain).
- If blood flow is interrupted for more than a few seconds, brain cells can die, causing infarction and damage.

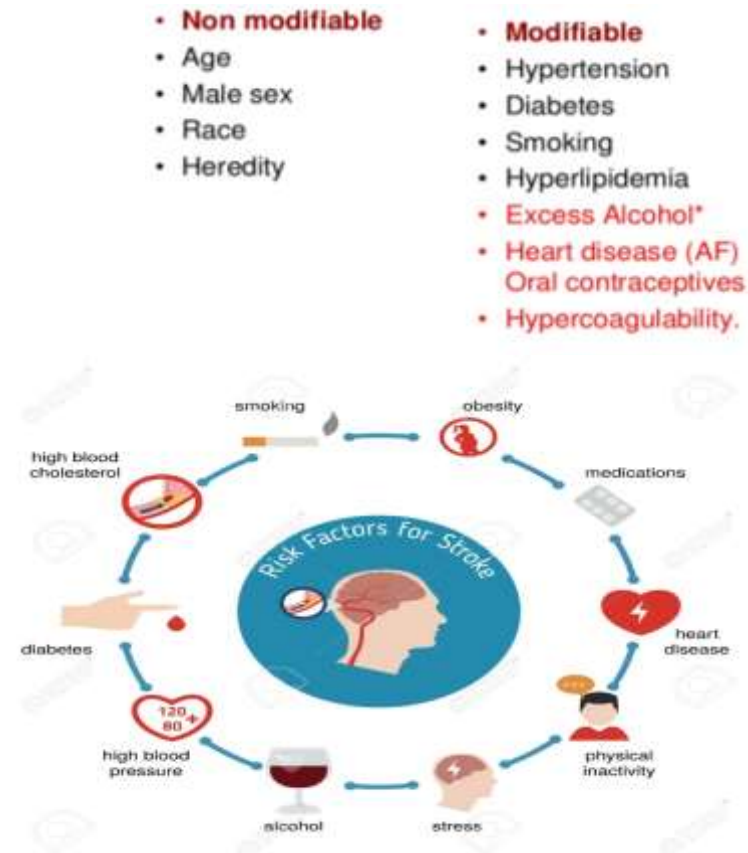


Cerebrovascular Stroke (CVS) or accident (CVA)

- Sudden or abrupt onset of focal neurological deficit **due to** occlusion of blood vessel (ischemic stroke) **or** ruptured vessel (hemorrhagic stroke) in the brain **causing** symptom and signs **lasting longer than 24 hours**.
- Stroke may preceded by transient ischemic attack (TIA) **characterized by:**
 - Focal neurological deficit
 - Lasts less than 60 min
 - Resolve completely within 24h.



- It is the 5th cause of death and 2nd leading **cause of death** and disability among old people.
- Each year, approximately 795,000 Americans experience a stroke, with 130,000 dying from stroke-related causes.
- **Predisposing or risk factors:**
 - Cardiac disease
 - Hypertension,
 - Hyperlipidemia
 - Diabetes
 - Smoking, alcohol and obesity



Types of stroke



- **Ischemic stroke (80%):**

- **Thrombosis:**

- **Cause:**

- Atherosclerosis, vasculitis (old age).

- Young patients:

- » Hypercoagulable states (eg, antiphospholipid antibodies, protein C deficiency, protein S deficiency, pregnancy)

- » Sickle cell disease

- » Substance abuse as cocaine



- **Embolism:**

- **Cause:**

- Atrial fibrillation (AF)

- Intracardiac thrombosis secondary to myocardial infarction (MI).

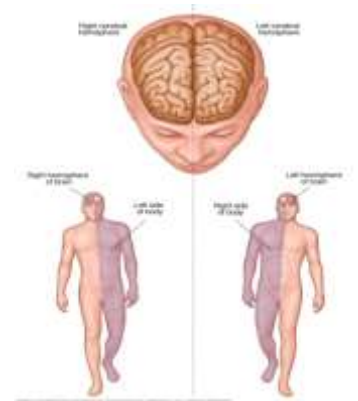
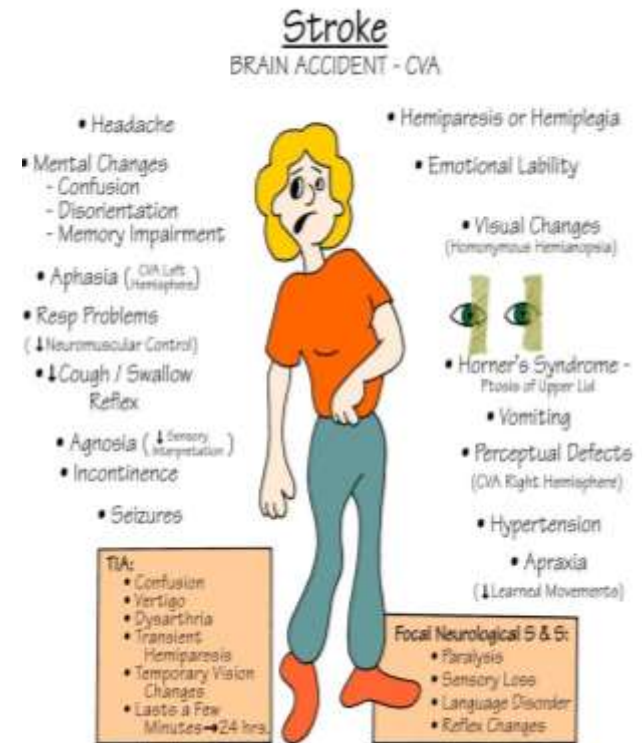
- Infective endocarditis.

- Fat, air emboli

- **Hemorrhagic stroke (20%):**
 - **Intra-cerebral hemorrhage:**
 - Causing increased pressure and damage to the surrounding cells.
 - **Causes:**
 - Hypertension, Hemorrhagic blood disease.
 - Drugs as anticoagulant (warfarin).
 - **Subarachnoid hemorrhage (SAH):**
 - **Cause:**
 - **Rupture** intracranial aneurysm or arterio-venous malformation.
 - **Characterized by:**
 - Sudden severe headache, vomiting, neck rigidity followed by disturbed consciousness.

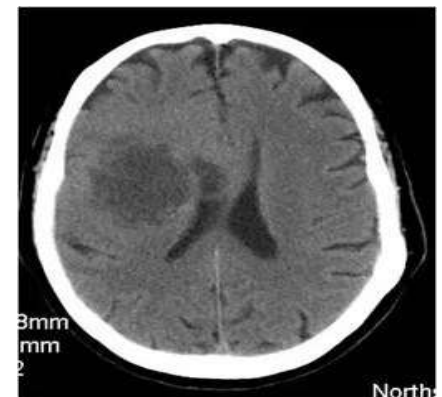
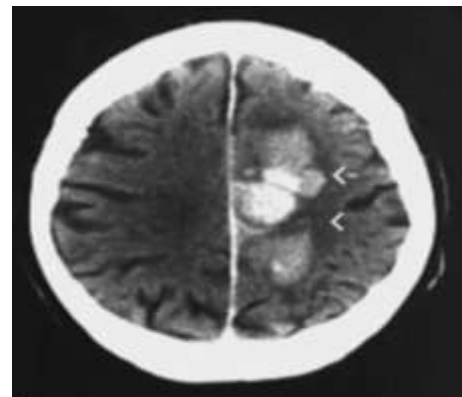
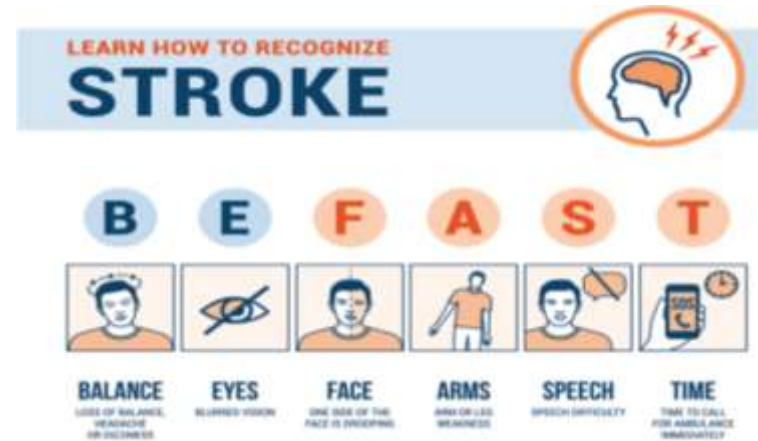
Signs and symptoms

- Disturbed conscious level.
- **Motor:** paralysis or paresis.
- **Sensory:** loss of sensation.
- Speech disorder (dysarthria or aphasia).
- Loss of vision.
- Cranial nerve palsy (facial nerve palsy or impaired swallowing).
- Seizures.
- Headache.
- Neck stiffness.



Diagnosis

- History and examination.
- Laboratory:
 - Lipid profile, blood glucose level
 - Cerebrospinal fluid analysis (CSF).
- Radiological:
 - CT brain (normal in early stages of ischemic stroke).
 - MRI brain.
 - ECG, echocardiography.



Stroke prevention

- **Primary stroke prevention (treat individuals with no previous history of stroke):**
 - **Antiplatelet aggregation as aspirin**
 - **Statins**
 - **Exercise**
 - **Lifestyle interventions (eg, smoking cessation, alcohol moderation)**
- **Secondary prevention (treat individuals who have stroke):**
 - **Antiplatelet aggregation**
 - **Antihypertensive**
 - **Statins**
 - **Lifestyle interventions**

Treatment

- **Assess** airway, breathing, circulation (ABC) and vital sign (blood pressure, Pulse, RR, temp.).
- **Supportive care:** including intensive care, life support and oxygen.
- **Nasogastric tube** if swallowing impaired to reduce the risk of aspiration pneumonia.
- **Care of skin** (to avoid bed sores) and care of bowel as catheterization/ bowel control.
- **Control** hypertension, diabetes and other risk factors.
- **Corticosteroids**, and antiseizure medications may be used

- **Thrombolytic** (risk of bleeding), **antiplatelet** as **aspirin**, and **anticoagulant** (AF patients) but there are strict criteria because the risk of bleeding.
- **Nimodipine** (calcium channel blocker) in SAH helps reduce intracranial vasospasm.
- **Surgical treatment** to clip or glue a leaking aneurysm, or evacuate a large intracerebral bleed.
- **Physiotherapy** to reduce muscle wasting and disused atrophy.

Dental aspect of cerebrovascular stroke

- Patients with stroke are at **high risk for poor or inadequate oral hygiene** which may affect their social status, communication, and quality of life.
- Preventive dental care **is most effective to prevent or reduce** oral disease including periodontal disease.
- Periodontal disease (Gingivitis, Periodontitis, Severe periodontitis) is associated with strokes (ischemic).
- Blood pressure monitoring and measurement **before dental procedures** and **should not exceed 140/90 mmHg.**



- **Regular cleaning, brush and flossing (once) teeth (dentures) after every meal (at least twice /day). A clean mouth will lead to:**
 - Improve oral hygiene status and reduce halitosis.
 - Prevent difficulties in eating, pain, or discomfort.
 - Reduce dental plaque and gingival bleeding.
 - Reduce the prevalence of oral opportunistic pathogens.
 - Increase the willingness to eat and contribute to a good nutritional intake and the removal of nasogastric tubes.

- The first dental appointment **should not be scheduled** until **6 months** after an initial stroke (except emergency).
 - **Maintain regular dental visits at least once/6 months.**
 - **Morning short stress- free appointments.**
 - Elective dental care should be postponed.
- Fixed appliances **are favored** over removable partial or complete dentures.
- **Avoid sugary drinks, snacks, food supplements, that can cause damage,** increase developing tooth decay.
- **Dry mouth** leading to difficulty of eating and drinking, also increase risk of infection (causing **bad taste and smell**), cavities and gum diseases.

- **Some stroke patients may experience tongue protrusion which can cause dry tongue, leading to thrush (candidiasis).**
- **Facial paralysis** is a common complication of stroke, which can **cause** difficulty speaking, eating, and drinking.
 - Also patients may have **difficulty brushing** or flossing, causing excessive infection, plaque buildup and may experience excessive tooth enamel wear and tear, increasing the risk of cavities.
- **Dentures can become loose or ill-fitting after a stroke which can cause gingivitis, leading to cheilitis and stomatitis. It is vital to adjusted or replaced it as needed to maintain good oral health.**

- If patient have swallowing problems (loss of gag reflex) or tube fed:
 - Ask your nurse show you the safest way to clean your teeth and oral cavity.
 - **Mouth care** at least 2 times/day by brushing the teeth with a suitable toothpaste and/or 0.12% chlorhexidine solution.
 - **Dentures** should be **clean regularly** and replaced if they do not fit well.
 - Electric toothbrush or adapted holders (patient with facial palsy).
 - Removal of excess secretions and application of lip balm.
 - Treat patient in **sitting upright position** **to prevent entering** foreign bodies to pharynx.



- **Patients received anticoagulants (warfarin), International Normalized Ratio (INR) testing should be done prior to dental treatment.**
 - INR should be between 1.5 and 3.5 for effective blood coagulation.
 - If it is less than 1.5, consult physician before treatment.
 - **Local hemostatic agents** should be available.
- **Avoid metronidazole and tetracyclines in patients taking warfarin, as the two medications decrease the effectiveness of warfarin and can increase INR.**

- **Effective pain control** during dental procedure and post-operative will reduce stress and risk for complications. Dental providers may administer **nitrous oxide with oxygen**.
- **Minimal amounts of anesthetic solution** should be used, with **low amounts of epinephrine**.
 - Norepinephrine avoided in cerebral hemorrhage (hypertension).
- **Any dentures or fixed appliances** should be taken out overnight and cleaned regularly.
- **Effective pain control**.
 - **Opioids avoided** in stroke, may cause severe hypotension.

Seizures (epilepsy)

- The brain normally contain millions of neuron interacting through electrical discharges.
- If there is disturbance in this interaction, it can cause changes in awareness, sensation, involuntary movements or behavior.

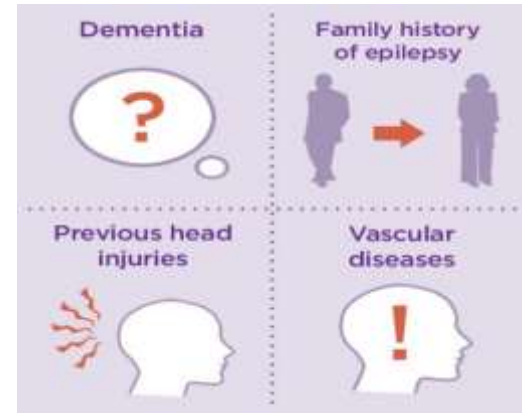


- **Seizure** is a paroxysmal uncontrolled electrical discharge of neurons in the brain that **interrupts normal functioning** (sensory, motor, autonomic with or without loss of consciousness) **leading to sudden, violent involuntary series of contraction of a group of muscle.**
- More prevalent in the young, mostly 5-20 year.
- It is associated with EEG changes.



Causes (risk factors) of epilepsy

- **Genetic**
- **Brain trauma:**
 - May affect any age, highest incidence in young adults.
 - It usually begin within 2 years after the injury.
- **Metabolic abnormalities:**
 - Hyperglycemia or hypoglycemia, electrolyte imbalances, and renal failure.
- **Brain tumors and other space-occupying brain lesions:**
 - May affect any age, more common after age 30 year.
 - Partial (focal) seizures most common initially
 - May progress to generalized tonic–clonic seizures



- **Other causes:**

- Alcohol use, or intoxication, or withdrawal
- Drugs: cocaine, anticonvulsants, barbiturates, benzodiazepines and opioids.

- **Cerebrovascular disease.**

- **Degenerative disorders:**

- Alzheimer disease: affect older people.

- **Infections:**

- Meningitis
- Encephalitis
- Brain abscess
- Acute severe systemic infections.

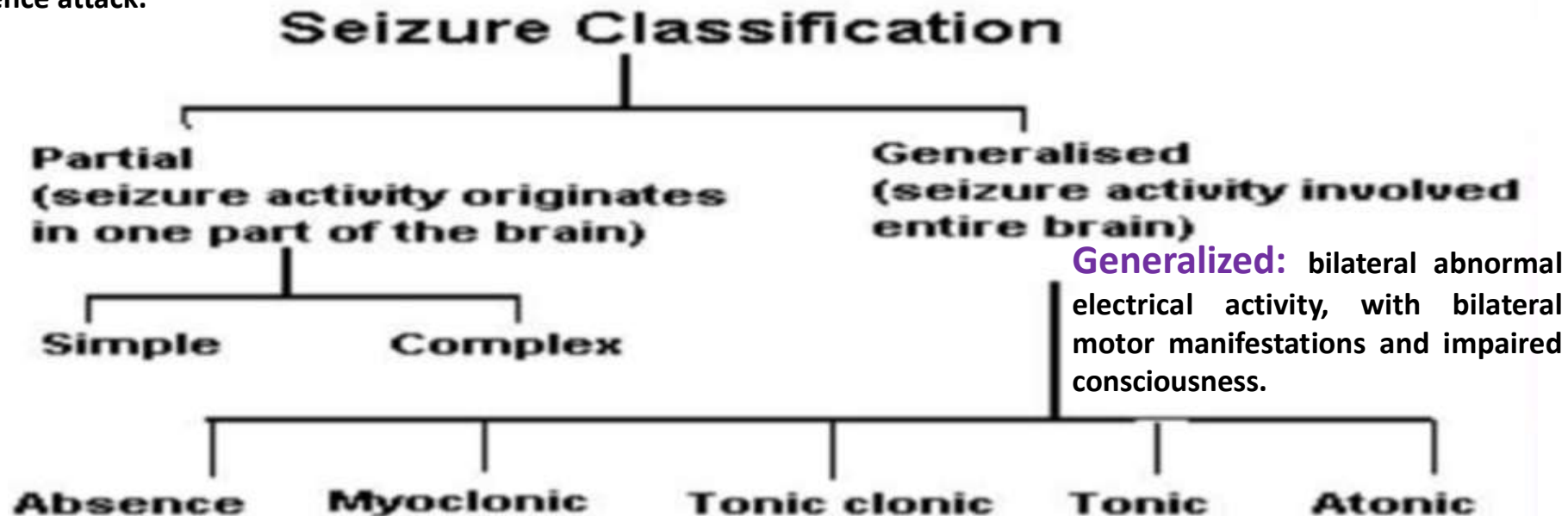


Classification

Partial (focal) seizure:

Simple- without loss of consciousness, e.g. jerking of a limb.

Complex- with loss of awareness, e.g. an absence attack.



Tonic-clonic seizures (grand mal seizures)

- Usually begins in childhood, or sometimes at puberty.

- It characterized by:

- **Pre-ictal (Aura):**

- **Warning sign:** mood change, irritability, hallucination or headache.

- **Ictal (seizure): sudden loss of consciousness** (10 minutes).

- **Tonic phase:** extension of limbs, rolling upward of the eyeball, cyanosis, clenching of the teeth, tongue bite and frothing from the mouth, retraction of the head to one side and urinary incontinence.
 - **Clonic phase:** whole muscles contract and relax repeatedly and rapidly.



- **Post –ictal:**

- Includes lassitude, headache, sleepiness, confusion or **Todd paralysis** (due to neuronal exhaustion).
- It last not more than 24 hour.

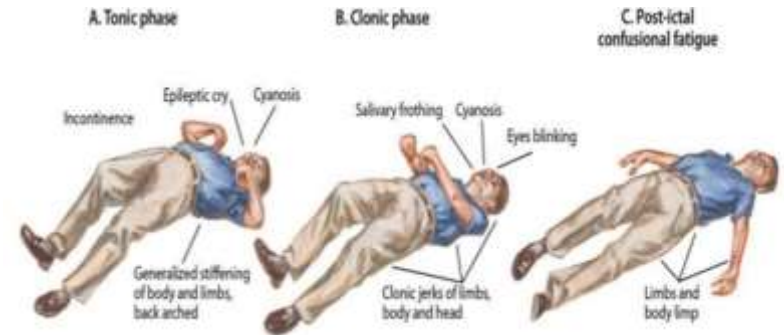
- **Treatment:**

- **Carbamazepine** (Tegretol), Gabapentin.

- **Phenytoin:** possible adverse effects:

- **Large doses:** nystagmus, ataxia and cerebellar damage.
- **Gingival swelling, thickening of the facial features.**
- **Allergy:** rashes or lupus- like reactions

- **Sodium valproate:** depresses numbers and function of the platelet causing **bleeding tendency.**



Petit mal (absence of seizures)

- It start in childhood and improve at puberty.
- No aura and no sequelae.
- **Characterized by:** Few second of
 - Staring look, blinking eyes or roll upward and eyelids twitch.
 - Sudden loss of consciousness (2-10 second).
 - Repeated many times if untreated.
- **Caused by** developmental abnormality and never due to acquired lesions such as tumors.
- May develop generalized tonic-clonic seizures in adult (known as primary generalized epilepsy).
- **Treatment:**
 - Valproate (depakin).
 - Ethosuximide (zarontin).

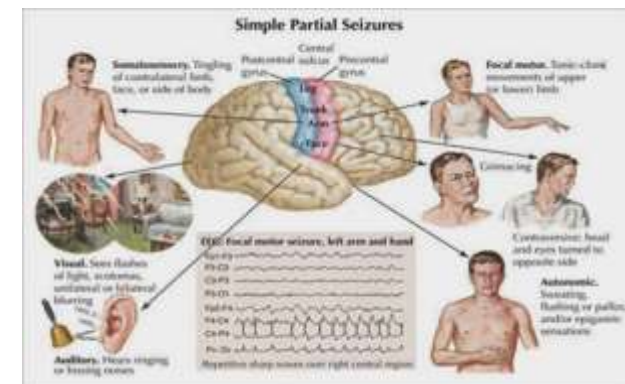


Other generalized types

- **Myoclonic seizures:**
 - Describe isolated muscle jerking.
- **Tonic seizures:**
 - Describe intense stiffening of the body not followed by convulsive jerking.
- **Atonic seizures:**
 - Sudden loss of tone, with falling and loss of consciousness.

Partial (Focal) epilepsy

- It characterized by recurrent, unprovoked seizures.
- It consist of:
 - **Simple seizures:**
 - Affect only temporal lobe and characterized by:
 - Motor (movement of part of limb, begin at the angle of mouth or in the hand).
 - Sensory (abnormal sensations paraesthesias).
 - Autonomic.
 - Special in the form of visual, olfactory or auditory hallucinations.
 - **No loss of consciousness.**
- **Treatment:** carbamazepine, valproate.

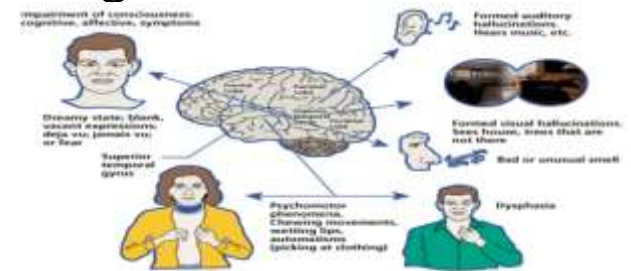


- **Complex (psychomotor epilepsy):**

- Originate in temporal lobe but can affect other region.

- Impaired consciousness.

- **Characterized by:**



- **Aura (few second):** in the form of olfactory or gustatory hallucination, emotion of fear, depersonalization, Déjà-vu phenomenon. Followed by:

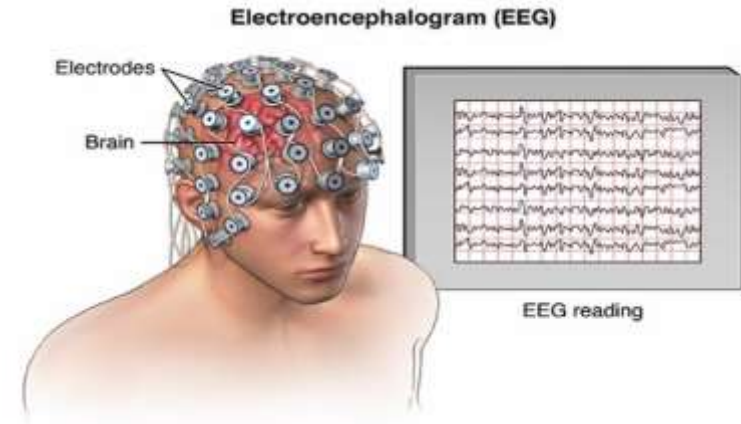
- **Absence:** the patient appears distant with staring eyes, may not respond to questions (altered consciousness).

- **Automatism:** may be champing of teeth, smacking and licking of lips or semi purposeful movement of limbs.

- **Amnesia** for the attack.

Diagnosis

- **History and examination.**
- **Laboratory:**
 - Serum electrolytes, blood sugar, CBC, liver and kidney function test.
 - Cerebrospinal fluid (CSF) study.
- **Radiological:**
 - Electroencephalogram (EEG).
 - Computed tomography (CT) and magnetic resonance imaging (MRI).
 - Positron emission tomography (PET).



Management

- **General:**

- Assess level of consciousness by Glasgow coma scale.
- Assess ABC.
- Monitoring of vital sign.
- Appropriate patient position.
- Mouth–gag and maintain oxygenation.
- Identification and avoidance of precipitating or trigger factors.
- **Blood sample for:**
 - Level of antiepileptic drugs, electrolytes, CBC, blood glucose, arterial blood gases (ABG).

Glasgow Coma Scale		
Response	Scale	Score
Eye Opening Response	Eyes open spontaneously	4 Points
	Eyes open to verbal command, speech, or shout	3 Points
	Eyes open to pain (not applied to face)	2 Points
	No eye opening	1 Point
Verbal Response	Oriented	5 Points
	Confused conversation, but able to answer questions	4 Points
	Inappropriate responses, words discernible	3 Points
	Incomprehensible sounds or speech	2 Points
	No verbal response	1 Point
Motor Response	Obeys commands for movement	6 Points
	Purposeful movement to painful stimulus	5 Points
	Withdraws from pain	4 Points
	Abnormal (spastic) flexion, decorticate posture	3 Points
	Extensor (rigid) response, decerebrate posture	2 Points
	No motor response	1 Point
Minor Brain Injury = 13-15 points; Moderate Brain Injury = 9-12 points; Severe Brain Injury = 3-8 points		

First Aid for Seizures

(Convulsions, generalized tonic-clonic, grand mal)



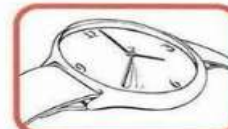
Cushion head, remove glasses



Loosen tight clothing



Turn on side



Time the seizure with a watch



Don't put anything in mouth



Look for I.D.



Don't hold down



As seizure ends...

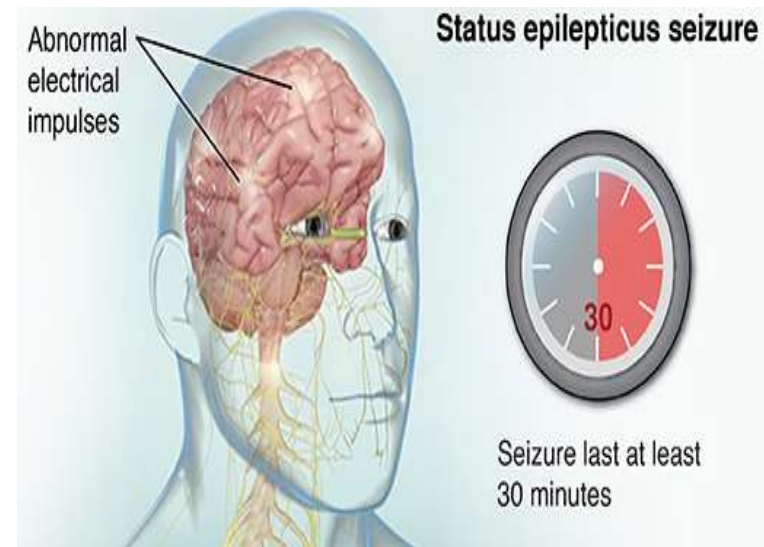


...offer help

- **Correct hypoglycemia, give i.v. glucose.**
- **Most seizures last only minutes and end spontaneously.**
- **Specific treatment:**
 - **Lorazepam 4-8 mg I.V. at 2 mg /min or**
 - **Diazepam 10-20 mg I.V. at 5 mg/min, or rectally according to route availability.**
- **Repeated every 5-10 minutes if seizure persist more than 10 min.**

Status epilepticus

- It defined as a seizure lasting longer than 30 minutes or repeated attack of generalized convulsions in between, the patient not regain consciousness.
- **Causes:**
 - Inappropriate withdrawal of antiepileptic drugs.
 - Others as stroke, CNS infection, alcohol, metabolic, traumatic and hypoxia.
- If the convulsion not stopped rapidly, coma deepens and death may occur.



Management

- **General:** as mentioned before.
- **Use I.V. 1st line drugs:**
 - **Diazepam** 10-20 mg i.v. at 5 mg/min and repeat once or **Lorazepam**.
 - If no I.V. access, give diazepam rectally or paraldehyde.
- **Repeat benzodiazepine if attacks persist.**
- **Use 2nd line drug:**
 - **Phenytoin or fosphenytoin.**
 - If the attack stop with I.V. diazepam to prevent recurrences.
 - Epanutin (phenytoin): IV infusion (contraindicated in cardiac arrhythmias, hypotension, recent MI)

- **3rd line of treatment:**

- I.V. Phenobarbital (if seizure persist after epanutin, valium).
- Ready to intubate the patient at this point.

- **If status persists > 90 minutes:**

- Use midazolam, thiopental or propofol anesthesia with assisted ventilation.

Dental aspect

- Keep dental chair as near to the floor as possible.
- The appointment was scheduled in the morning hours.
- Stress is a factor that provokes seizures, so ensured that were eliminated before the start of treatment.
- If patient has any fluid or dental instrument in their mouth, should be rolled the patient onto their side to help the patient breathe and prevent aspiration of foreign objects or secretions.
- Anything around the neck, like a necktie, scarf should be loosened.
- Eyeglasses must be removed.

- **A dentist evaluate an epileptic patient for any signs of gingival hyperplasia or dry mouth, and they will share the following oral hygiene steps to help them prevent tooth decay and gum disease:**
 - **Floss daily.**
 - **Brush frequently throughout the day and at bedtime with fluoride toothpaste.**
 - **Maintain a healthy diet and get adequate nutrition.**
 - **Use mints or lozenges to stimulate saliva production and prevent cavities.**

- Epileptic seizures usually lead to minor oral injuries, such as tongue biting.
- If experiencing an epileptic seizure:
 - Remove all dental instruments or any harmful things away from the patient to avoid:
 - Fractures or dislocation of vertebrae or limbs.
 - Face Injuries as lacerations, hematomas, fractures of the facial skeleton, subluxation or loss of teeth.
 - Temporomandibular joint (TMJ) subluxation, tooth injury, tongue biting, lacerations of lips or buccal mucosa.
- Check what time the seizure started- Call emergency if the seizure lasts longer than 5 minutes.

- **Once the seizure has finished:**
 - Gently move them into the recovery position.
 - Do not undertake further **dental treatment that day.**
 - **Do not allow the patient to leave** until full recovery.
 - Contact the patient's family, **if he or she is alone.**
 - Evaluate the level of consciousness and examine his or her injuries.
 - Do not give them anything at all to eat and drink **until fully recovered.**

- **Patient on phenytoin, phenobarbital and carbamazepine can be at risk of fracture** due to alter the metabolism and clearance of vitamin D.
- **Aspirin, azoles and metronidazole** can interfere with phenytoin, also increase the bleeding tendency induced by valproate.
- Propoxyphene and erythromycin can **interfere with carbamazepine.**
- **Large doses of lidocaine** given intravenously for severe arrhythmias may occasionally **cause convulsions.**
- **Acrylic** is probably **better used** for prostheses than **porcelain.**

- A 68 year old hypertensive patient came to dental clinic for follow up. He developed weakness in his left side of the body. CT brain showed white lesion in right cerebral hemisphere. What is your diagnosis?
- A. Cerebral hemorrhage.
- B. Cerebral infarction.
- C. Vasovagal syncope.
- D. Grand-mal epilepsy.

- A 72- year- old male patient presented by toothache. He has recent history of cerebrovascular stroke with bulbar palsy. What is the best management in this case?
- A. Keep patient in supine and raising his legs.
- B. Treat patient in upright position.
- C. Anticoagulant given before manipulation.
- D. Tricyclic Antidepressants.

- A 65 year old hypertensive male patient presented to dental outpatient clinic for follow up. He develops weakness in his left side of the body associated with numbness and tingling sensation. The condition completely improved within one hour. What is your diagnosis?
- a) Transient ischemic attack
- b) Cerebrovascular stroke.
- c) Reversible ischemic neurological deficit.
- d) Complex (psychomotor) partial seizures.

- What is the most important investigation for diagnosis of epilepsy?
- A. EEG
- B. ECG
- C. Blood chemistry
- D. C T Brain

- Which of the following IV drugs is effective in treatment of status epilepticus?
- a) Cabamazipine.
- b) Gapapentin.
- c) Diazepam.
- d) Phenytoin.

- Which of the following disorders presented with abnormal movement and aura?
- A. Petit mal epilepsy.
- B. Migraine headache.
- C. Grand mal seizures.
- D. Cerebrovascular stroke.



Thank You